Isolation means:

One container cannot interfere with another container or the host system.

1. **What is Docker?**
   Docker is a containerization platform that packages an application and its dependencies into a single container that runs consistently across environments.
2. **Containers vs Virtual Machines (VMs)?**
   Docker containers share the host OS kernel, are lightweight & start faster, while VMs include separate OS instances.
3. **What are images and containers?**
   - *Image:* Read-only template
   - *Container:* A running instance of an image.
4. **What is a Dockerfile?**
   A text file with instructions to build a Docker image (base image, commands, ENV variables).
5. **What is Docker Compose?**
   A tool to define and run multi-container applications with a docker-compose.yml.

🔥 Intermediate Docker Questions

6. **Explain Docker networking.**
   Docker networks enable containers to communicate via bridge, host, or overlay networks.
7. **What are Docker volumes?**
   Volumes store persistent data that containers can share or preserve across restarts.

8. **How to optimize Docker images?**
   Use minimal base images, multi-stage builds, fewer layers, and clean caches.

9. **Difference between CMD and ENTRYPOINT?**
   ENTRYPOINT sets the executable; CMD provides default arguments.

10. **Explain container isolation.**
    Docker uses namespaces (for processes) and cgroups (for resource limits).

💡 Advanced Docker Questions

11. **What is Docker Swarm?**
    Native Docker orchestration for clustering containers.

12. **How is container orchestration handled?**
    Tools like Kubernetes manage scaling, load balancing, and availability.

13. **How do you secure Docker containers?**
    Run with least privilege, use trusted images, scan images for vulnerabilities, apply AppArmor/SELinux policies.

14. **What is multi-stage build?**
    Reduces final image size by separating build and runtime stages.

15. **Scenario:** A Docker image is too large — how do you reduce it?
    Remove unnecessary tools, use smaller base, multi-stage builds, minimize layers.

1️⃣ What is Docker Networking (Simple Definition)

Docker networking allows containers to talk to:

- Other containers
- The host machine
- The outside world (internet)

Each container has:

- Its own **IP**
- Its own **network namespace**

Docker creates **virtual networks** to manage communication.

2️⃣ Types of Docker Networks (With Clear Use Cases)

🔹 1. Bridge Network (MOST COMMON)

📌 **Default network**

- Containers can talk to each other using **container name**
- Used for **single-host applications**

Example:

docker network create my-network

👉 Best for:

- Backend + DB containers
- Local development
- Microservices on one server

🔹 2. Host Network

- Container shares **host's network**
- No isolation
- Faster, but **less secure**

👉 Used when:

- Performance is critical
- Legacy apps need host networking

🔹 **3. Overlay Network**

- Used in **Docker Swarm / Kubernetes**
- Enables communication across **multiple hosts**

👉 Used in:

- Production clusters
- Distributed microservices

🔹 **4. None Network**

- No network access
- Fully isolated container